

CHANDNI VERMA

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👤 PROFILE

Remote Sensing and Geospatial Professional with expertise in SAR interferometry, satellite image processing, DEM analysis, Python-based geospatial development, and automated Earth observation workflows.

🏢 PROFESSIONAL EXPERIENCE

Junior Research Fellow

09/2024 – 03/2026 | Chandigarh, India

CSIR-Central Scientific Instruments Organisation (CSIO)

A comprehensive Geomorphic Anomaly Investigation over Sutlej Basin, India

- Developed **GeoProfiler**, an automated Python-based geospatial tool for extracting topographic line and swath profiles from DEM datasets, enabling scalable terrain analysis and generation of publication-quality vector outputs for geomorphological investigations.
- Conducted DEM-based geomorphic, hydrological, and morphotectonic analyses using multi-source geospatial datasets and computational tools including Python, MATLAB, TopoToolbox, and GIS platforms.
- Performed watershed and drainage-network delineation, fluvial terrace mapping, longitudinal river-profile analysis, basin asymmetry assessment, and χ -index-based geomorphic analysis to investigate landscape evolution and tectonic influences in the Himalayan terrain.
- Conducted field investigations across Himachal Pradesh for validation of mapped geomorphic features, including terrace verification, elevation measurements, lithological logging, sediment characterization, geomorphic documentation, and soil-sample collection for OSL dating.
- Prepared technical figures, geomorphic maps, and visualizations, and completed detailed project reports and documentation.

Research Trainee - Advanced SAR Data Processing for Ground Surface Movement

10/2024 – 10/2024 | Dehradun, India

Indian Institute of Remote Sensing (IIRS), ISRO

- Configured Linux-based computational environments and preprocessing workflows for SAR interferometric analysis.
- Performed guided Differential SAR Interferometry (DInSAR) and Small Baseline Subset (SBAS) processing workflows using ASF, SNAP, and ISCE for ground-surface deformation analysis.
- Acquired practical exposure to SAR time-series analysis frameworks, including MintPy, for deformation monitoring and interferometric interpretation.

🎓 EDUCATION

M.Tech in Remote Sensing and GIS (Geosciences)

08/2022 – 07/2024 | Dehradun, India

Indian Institute of Remote Sensing (IIRS), ISRO

M.Sc in Geology

07/2014 – 06/2016 | Chandigarh, India

Panjab University

B.Sc in Geology

07/2011 – 06/2014 | Chandigarh, India

Panjab University

Minors: Mathematics and Chemistry

📁 PROJECTS

Land Subsidence Characterization in Northwest India in relation to Groundwater Over-extraction, Hydrogeology, and Active Tectonics

DInSAR | PS-InSAR | Sentinel-1 | GRACE | Aquifer System Compaction

- Analyzed GRACE-derived groundwater storage anomalies to investigate regional groundwater depletion trends.
- Processed and analyzed 151 Sentinel-1 SLC scenes (2018–2022) using **DInSAR** and **PS-InSAR** techniques in ENVI/SARscape and SARPROZ for time-series analysis of ground deformation and land subsidence.
- Performed aquifer-system compaction modeling using piezometric groundwater-level data and compared modeled deformation with PS-InSAR-derived displacement trends.
- Conducted field investigations at subsidence-affected locations to validate deformation patterns through on-site observations and photographic documentation of structural damage in affected buildings.

GeoProfiler: Automated Topographic Profile Extraction and Terrain Visualization

Python | Streamlit | Rasterio | GeoPandas | NumPy | SciPy | Matplotlib

- Developed and deployed GeoProfiler, an open-source Python web application for automated topographic line and swath profile extraction, DEM visualization, and terrain analysis.
- Implemented DEM visualization with hillshade rendering, automatic CRS handling, GeoJSON/Shapefile support, and export to PNG, PDF, and CSV formats.

- Designed an intuitive browser-based workflow that streamlines terrain analysis and reduces repetitive GIS processing for geomorphological and topographic investigations.
- Published the application as a publicly accessible web platform with integrated documentation, sample datasets, and interactive profile generation capabilities.

Web App: geoprofiler.streamlit.app 

GitHub: github.com/chandnivermageo/GeoProfiler-WebApp 

SAR-Based Reservoir Sedimentation and Storage Capacity Assessment using Sentinel-1 GRD Imagery

Sentinel-1 | Google Earth Engine

- Conducted SAR-based reservoir sedimentation assessment using Sentinel-1 GRD imagery and multi-temporal water-spread analysis in Google Earth Engine (GEE) and ArcGIS.
- Integrated SAR-derived water-spread extent with India WRIS reservoir-level data to estimate revised storage capacity and sedimentation impacts.

Snow-Cover Delineation and Change Detection using Landsat 8 and DEM Integration

Python | Landsat 8 | DEM

- Developed an automated Python-based workflow for snow-cover delineation and change detection using a multi-criteria classification approach integrating spectral indices, radiometric thresholds, elevation filtering, and morphological cleanup for improved snow extraction accuracy.
- Performed raster-based spatial analysis, binary change mapping, and elevation-stratified snow distribution assessment using Rasterio, NumPy, SciPy, and Matplotlib.

GitHub: github.com/chandnivermageo/Snow-Cover-Change-Detection 

TECHNICAL SKILLS

Remote Sensing & Earth Observation: Optical and SAR image processing, DInSAR, PS-InSAR, SAR time-series analysis, change detection, snow and water-body delineation.

Terrain & Geomorphic Analysis: DEM processing, terrain analysis, watershed and drainage delineation, Topographic profile extraction, fluvial geomorphology, terrace mapping, morphotectonic analysis.

Programming & Scientific Computing: Python, Streamlit, MATLAB, R, Rasterio, NumPy, SciPy, GeoPandas, Matplotlib, TopoToolbox, geospatial workflow automation.

SOFTWARE & TOOLS

ArcGIS • QGIS • Google Earth Engine • ENVI/SARscape • SNAP • SARPROZ • ERDAS IMAGINE • Google Earth Pro

WORKSHOP & CONFERENCES

- Poster Presentation, IIRS Academia Meet (IAM-2024) - *Land Subsidence Characterization in NW India using GRACE and SAR Data*
- Delivered oral poster presentation at 10th Chandigarh Science Congress (CHASCON-2016)
- Abstract Selected, 6th World Congress on Disaster Management (2023): *Reservoir Sedimentation Assessment of Tehri Reservoir Using SAR Data*
- Attended workshop on National Aquifer Mapping (NAQUIM) and Groundwater Data Dissemination for Sustainable Management, CGWB, March 2026.
- Attended International Workshop on Observing Ground Deformation from Space and in the Field, CSIR-CSIO, December 2024.

LANGUAGES

Hindi, English, and Punjabi

ACHIEVEMENTS AND SCHOLARSHIP

- GATE 2025 (GE-Geomatics): All India Rank 153, GATE Score 496
- GATE 2024 (GE-Geomatics): All India Rank 239, GATE Score 445
- Recipient AICTE M.Tech GATE Scholarship (2022-2024)
- GATE 2022 (GG-Geology & Geophysics): All India Rank 763, GATE Score 370

CO-CURRICULAR AND EXTRA-CURRICULAR ACTIVITIES

- Winner (Mixed Table Tennis & Badminton), CSIR-CSIO Sports Tournament, 2025
- Key team member in organizing the International Conference on Land Surface Deformation at CSIR-CSIO (Dec 2024)
- Held leadership roles as General Secretary (2015-16) and Cultural Secretary (2013-14) at Pick & Hammer Club, Panjab University

INTERESTS

Singing, Guitar, Badminton and Table Tennis